SlipSurfaceIQ LE

DXF Import Guide

Bring slope cross-sections from CAD into SlipSurfaceIQ LE

Version 2026.7.0

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01 Overview

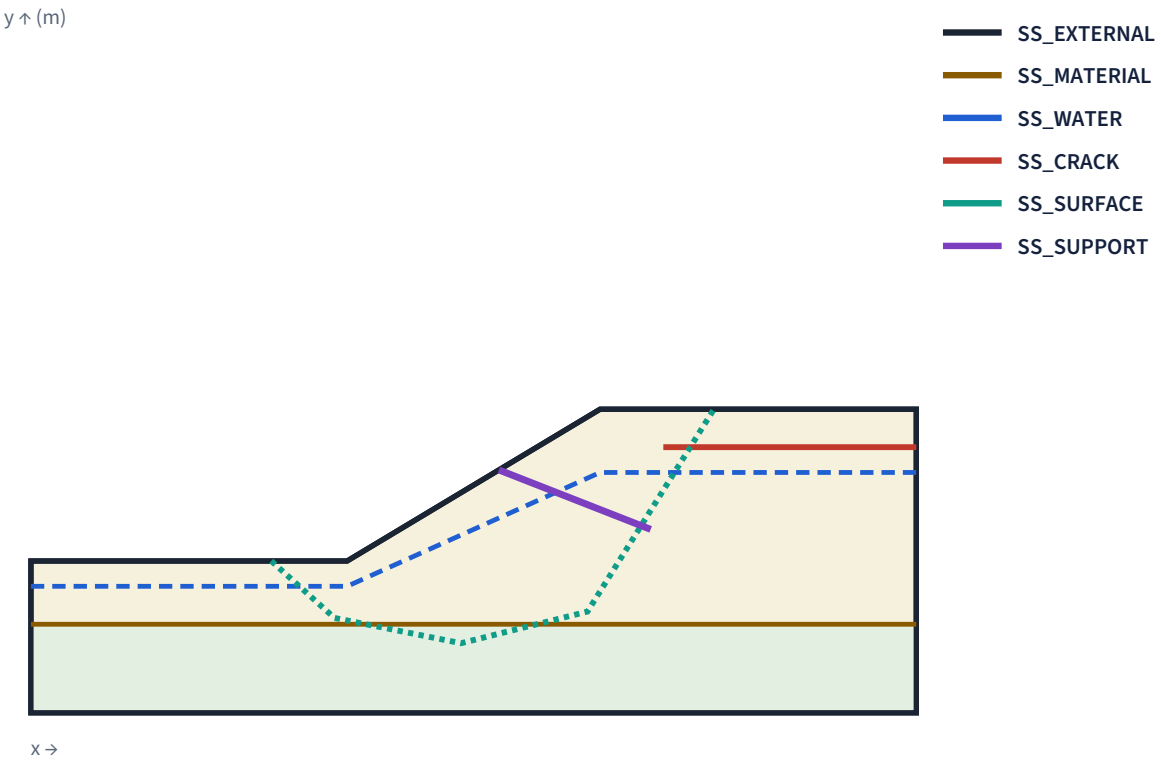
File → Import DXF reads the 2D geometry of a slope from a CAD drawing. Each kind of line is recognised by its layer name.

• **Geometry only.** The external boundary, material boundaries, water table, tension crack and a user slip surface are read. Materials, loads, supports and settings are not part of DXF.

• **Replaces the open model.** Importing creates a new model with one default material (Soil 1) and the file name as title. **Ctrl+Z** brings the previous model back.

• **Checked on import.** The geometry is built immediately; any problem is reported and nothing changes. Skipped entities and layers are listed in the message after a successful import.

• **File → Export DXF** writes the same layers, so a model can go to CAD and back.



The layers of a slope model. The same model is in the template drawing **docs/DXF_Layer_Template.dxf**.

02 Layer reference

Layer names are not case-sensitive (ss_water = SS_WATER).

Layer	Entities	Becomes	Rules
`SS_EXTERNAL`	exactly 1, closed	External boundary	Required. One closed, non-self-intersecting polyline.
`0`	1 closed	External boundary	Used when SS_EXTERNAL is not used: a closed polyline on layer 0 counts as the outline. Open entities on layer 0 are skipped.
`SS_MATERIAL`	any number	Material boundaries	Open lines that split the domain into regions; ends must touch the outline (see chapter 05).
`SS_WATER`	at most 1	Water table	Left to right with increasing x; extended horizontally to both model ends.
`SS_CRACK`	at most 1	Tension crack line	Its depth below the ground surface is the crack depth.
`SS_SURFACE`	at most 1	User slip surface	Stored as the user-defined surface; select it in Surface Options.
`SS_SUPPORT`	export only	—	Written on export (one line per support); skipped on import.
`any other`	—	skipped	Listed in the import message.

Start from the template
Open **docs/DXF_Layer_Template.dxf** in your CAD program: it already has every layer and one entity on each. Draw over it, delete what you do not need, and save as DXF.

03 Entities, curves and units

What is read from the drawing and how it is converted to metres.

Entities

Entity	Read as
`LINE`	Two-point line.
`LWPOLYLINE`	Polyline; arc segments (bulges) are converted to straight pieces.
`POLYLINE`	2D polyline (same as LWPOLYLINE).
`ARC`	Converted to straight pieces.
`SPLINE`	Converted to straight pieces.
`CIRCLE, ELLIPSE, TEXT, MTEXT, INSERT (blocks), HATCH, DIMENSION, POINT, …`	Skipped and listed in the import message.

Curves are replaced by straight segments that stay within **1 cm** of the true curve. Blocks are not exploded: explode them in CAD first.

Units

The drawing unit ($INSUNITS, set with the UNITS command in most CAD programs) is converted to metres. A drawing without units is read as metres, with a warning.

Drawing unit	Factor to metres	Example: 12 500 drawing units
`Millimetres`	0.001	12.5 m
`Centimetres`	0.01	125 m
`Metres`	1	12 500 m
`Feet`	0.3048	3 810 m
`Inches`	0.0254	317.5 m
`Unitless`	1 (warning)	12 500 m

Coordinates
x is horizontal and y is vertical (elevation), both in the drawing plane at Z = 0. A cross-section drawn in a rotated UCS or at an elevation Z must be moved to the world XY plane first. Geometry with Z ≠ 0 is refused.

04 Preparing a drawing in CAD

A checklist that avoids nearly every import error.

#	Step	Typical commands
1	Copy the cross-section into a new drawing.	WBLOCK, COPYCLIP
2	Set the drawing units (preferably metres).	UNITS, INSUNITS
3	Explode blocks and groups.	EXPLODE
4	Move everything to Z = 0 in the world XY plane.	FLATTEN, UCS World
5	Join the outline into ONE polyline and close it.	JOIN, PEDIT → Join / Close
6	Put each line on its layer (SS_EXTERNAL, SS_MATERIAL, …).	LAYER, Properties
7	Join the water table into one polyline running left to right.	JOIN, REVERSE
8	Extend material boundaries until they meet the outline.	EXTEND, object snaps
9	Delete hatches, text and dimensions (or keep them on other layers).	ERASE
10	Save as DXF (any version, e.g. R2010).	SAVEAS → DXF

05 Layer details

External boundary — SS_EXTERNAL

One closed outline of the whole model: ground surface on top, the sides and the base. The vertex order does not matter. Separate LINE entities are not joined automatically; the import stops with “SS_EXTERNAL must be a closed polyline.”

Material boundaries — SS_MATERIAL

Each open line divides the region it crosses. Line ends lying close to the outline or to another boundary are snapped onto it. After the import, assign materials with **Properties → Assign Material**; until then every region uses Soil 1.

> **Ends must reach the outline**
> Ends are snapped within **0.2 % of the larger model dimension** (12 cm for a 60 m wide model). A boundary that stops farther away is kept but does not split the region; the import message then says “Material boundary N does not divide a region”, and the Modeler status bar repeats it. Extend the lines onto the outline in CAD.

Water table — SS_WATER

One polyline with increasing x (left to right, no back-steps). It is extended horizontally from its first and last points to the model ends. After import the groundwater method is *water table*; per-material Hu is set in the material dialog.

Tension crack — SS_CRACK

One line or polyline above or on the ground; at least two points with different x. The crack depth at each x is its distance below the ground surface. Set the water filling in **Project Settings → Groundwater**.

User slip surface — SS_SURFACE

One polyline, arc or spline. It is stored as the user-defined surface, but the surface type is not switched: choose **Surfaces → Surface Options → User-defined** to analyse it.

06 After the import

Complete the model in the Modeler.

1. Read the import message: it lists skipped entities and layers.

2. Define materials (**Properties → Define Materials**) and assign them to the regions.

3. Add loads, supports and facings, and set seismic coefficients if needed.

4. Choose the slip surface type and search options.

5. Compute (F5), then save the model as **.slp.json** — that file keeps everything, DXF keeps only geometry.

07 Exporting to DXF

File → Export DXF writes an AutoCAD R2010 drawing in metres with the layers below. It needs an external boundary.

Written	Not written
SS_EXTERNAL (closed), SS_MATERIAL, SS_WATER, SS_CRACK, SS_SURFACE, SS_SUPPORT (single supports)	Materials and assignments, loads, bolt patterns, facings, notes, settings and results

Exported files import back unchanged (the supports layer is skipped on import), so DXF is a safe way to edit the geometry in CAD and return.

08 DXF from Python

The import and export functions can be used in a Python script (Tools → Python scripting) or any Python program.

import_in_a_script.py

```python
from slipsurface.exchange import import_dxf

project, warnings = import_dxf(TEMPLATE)   # path of a .dxf file
for line in warnings:
    print(line)
project.materials[0].c = 10   # continue the model as usual
project.materials[0].phi = 28
```

In the scripting editor write the file path instead of `TEMPLATE`, e.g. `r"C:\models\section.dxf"`.

write_a_dxf_with_ezdxf.py

```python
import ezdxf
from pathlib import Path
from slipsurface.exchange import import_dxf

doc = ezdxf.new("R2010")
doc.units = ezdxf.units.M
msp = doc.modelspace()
for layer in ("SS_EXTERNAL", "SS_MATERIAL", "SS_WATER"):
    doc.layers.add(layer)
outline = [(0, -10), (60, -10), (60, 10), (40, 10), (20, 0), (0, 0)]
msp.add_lwpolyline(outline, close=True, dxfattribs={"layer": "SS_EXTERNAL"})
msp.add_line((0, -4), (60, -4), dxfattribs={"layer": "SS_MATERIAL"})
msp.add_lwpolyline([(0, -2), (20, -2), (40, 5), (60, 5)],
                   dxfattribs={"layer": "SS_WATER"})
path = Path(TMP) / "section.dxf"
doc.saveas(path)
project, warnings = import_dxf(path)
print(len(project.build_geometry().regions), "regions")
```

Export works the same way: `from slipsurface.exchange import export_dxf` and `export_dxf(project, "section.dxf")`.

09 Messages and fixes

Import messages, word for word, and what to do.

Message	Cause and fix
`DXF needs exactly one closed SS_EXTERNAL (or layer 0) boundary.`	No external boundary, or more than one closed polyline on SS_EXTERNAL / layer 0. Keep one closed outline; move other closed shapes to another layer.
`SS_EXTERNAL must be a closed polyline.`	The outline is open, or drawn as separate LINEs. Join it into one polyline and close it (JOIN / PEDIT → Close).
`Multiple entities on SS_WATER; join them before importing.`	SS_WATER, SS_CRACK or SS_SURFACE holds more than one entity. Join them into one polyline.
`Only planar XY geometry at Z=0 is supported.`	Some vertices have Z ≠ 0. Flatten the drawing (FLATTEN, or set Z to 0).
`The external boundary intersects itself (invalid polygon).`	The outline crosses itself. Fix the vertex order or remove the loop.
`The external boundary has zero area.`	The outline is degenerate (all points on a line).
`Water table points must be in increasing x order.`	The water table runs right to left or turns back. Reverse it (REVERSE) and remove back-steps.
`The tension crack line needs at least 2 points with different x values.`	The crack line is a single point or vertical.

Notice after a successful import	Meaning
`DXF has no units; coordinates interpreted as metres.`	Set the drawing units if the drawing is not in metres.
`Skipped entities/layers: …`	These entities were ignored (see chapter 03).
`Material boundary N does not divide a region (… m of it is loose): …`	Its ends do not reach the outline or another boundary; extend them (chapter 05).
`DXF transfers geometry only. Assign materials, loads and supports separately.`	Always shown; continue with chapter 06.